Section 7.3 Exercises

Directions: For exercises 1 - 4, find the exact values of a) $\sin(2x)$, b) $\cos(2x)$, and c) $\tan(2x)$ without solving for x.

1. If $\sin x = \frac{1}{8}$, and x is in quadrant 1.

2. If $\cos x = \frac{2}{3}$, and x is in quadrant 1.

3. If $\cos x = -\frac{1}{2}$, and x is in quadrant 3.

4. If $\tan x = -8$, and x is in quadrant 4.

Directions: For exercises 5 - 6, find the values of the six trigonometric functions using the given information.

5. $\cos(2\theta) = \frac{3}{5}$ and $90° \leq \theta \leq 180°$

6. $\cos(2\theta) = \frac{1}{\sqrt{2}}$ and $180° \leq \theta \leq 270°$

Directions: For exercises 7 - 8, simplify and find the exact value for:

7. $2\sin\left(\frac{\pi}{4}\right)\ 2\cos\left(\frac{\pi}{4}\right)$

8. $4\sin\left(\frac{\pi}{8}\right)\ \cos\left(\frac{\pi}{8}\right)$

Directions: For exercises 9 - 14, find the exact value using half-angle formulas.

9. $\sin\left(\frac{\pi}{8}\right)$

10. $\cos\left(-\frac{11\pi}{12}\right)$

11. $\sin\left(\frac{11\pi}{12}\right)$

12. $\cos\left(\frac{7\pi}{8}\right)$

13. $\tan\left(\frac{5\pi}{12}\right)$

14. $\tan\left(-\frac{3\pi}{8}\right)$

Directions: For exercises 15 - 18, find the exact values of a) $\sin\left(\frac{x}{2}\right)$, b) $\cos\left(\frac{x}{2}\right)$, and c) $\tan\left(\frac{x}{2}\right)$ without solving for x.

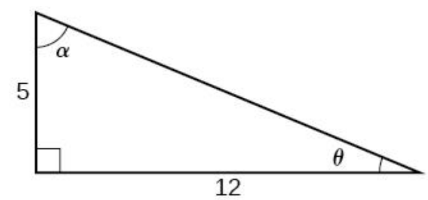
15. If $\tan x = -\frac{4}{3}$, and x is in quadrant IV.

16. If $\sin x = -\frac{12}{13}$, and x is in quadrant III.

17. If $\csc x = 7$, and x is in quadrant II.

18. If $\sec x = -4$, and x is in quadrant II.

Directions: For the exercises 19 - 22, use the right triangle below to evaluate the requested trigonometric expressions.



19. Find $\sin(2\theta), \cos(2\theta)$, and $\tan(2\theta)$.

20. Find $\sin(2\alpha)$,$\cos(2\alpha)$, and $\tan(2\alpha)$.

21. Find $\sin\left(\frac{\theta}{2}\right)$, $\cos\left(\frac{\theta}{2}\right)$, and $\tan\left(\frac{\theta}{2}\right)$.

22. Find $\sin\left(\frac{\alpha}{2}\right)$, $\cos\left(\frac{\alpha}{2}\right)$, and $\tan\left(\frac{\alpha}{2}\right)$.

Directions: For exercises 23 - 28, simplify each expression. Do not evaluate.

23. $\cos^2(28^\circ) - \sin^2(28^\circ)$

24. $2\cos^2(37^\circ) - 1$

25. $1 - 2\sin^2(17^\circ)$

26. $\cos^2(9x) - \sin^2(9x)$

27. $4\sin(8x)\cos(8x)$

28. $6\sin(5x)\cos(5x)$

Directions: For exercises 29 - 32, prove the identity given.

29. $(\sin t - \cos t)^2 = 1 - \sin(2t)$

30. $\sin(2x) = -2\sin(-x)\cos(-x)$

31. $\cot x - \tan x = 2\cot(2x)$

32. $\frac{1+\cos(2\theta)}{\sin(2\theta)}\tan^2\theta = \tan\theta$

Directions: For exercises 33 - 36, rewrite the expression with an exponent no higher than 1.

33. $\cos^2(5x)$

34. $\cos^2(6x)$

35. $\sin^4(8x)$

36. $\sin^4(3x)$